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The function y = k / x — properties and graph

Inverse proportion: the first function of the course whose graph is not a straight line, and whose defining feature is a product that never changes.

LESSON 25–27 | 3 × 40 MIN | ALGEBRA 8, §7 | STAGE 9 · 10.1–10.2, 11.2

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Recognise inverse proportion and find k from one point.
- Draw the graph of $y = k/x$ by table of values.
- State the properties: domain, the two branches, quadrants, symmetry, asymptotes.
- Solve practical problems in which a product is constant.

TEXTBOOK

National	§7, pp. 34–38
Cambridge	Learner's Book pp. 213–224, 242–249
Workbook	Workbook 10.1, 11.2

01 Explanation

Inverse proportion

DEFINITION

Two quantities are **inversely proportional** if their product is constant. Writing that constant as k, we get $xy = k$, that is

$$y = \frac{k}{x}, \quad k \neq 0, x \neq 0$$

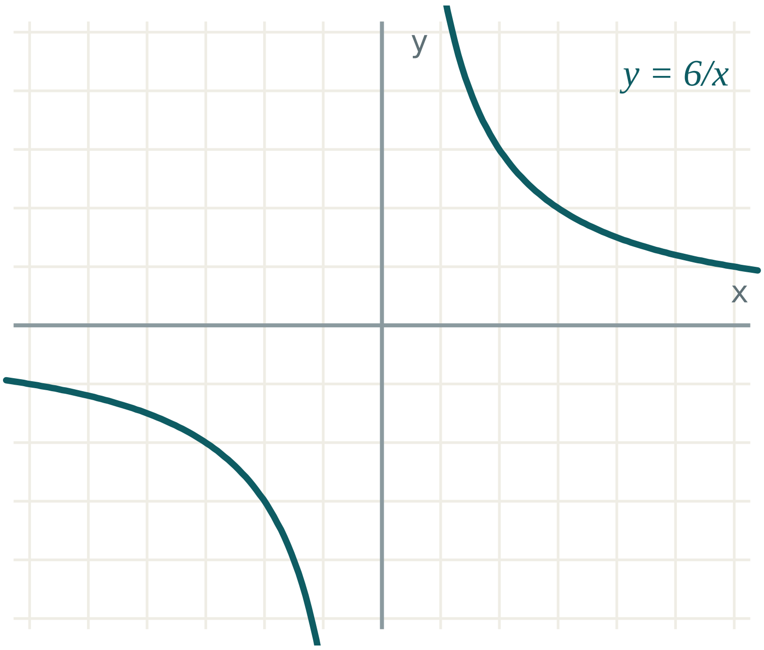
Everyday examples: a fixed distance covered at different speeds ($t = \frac{s}{v}$); a fixed amount of work shared between different numbers of workers; a rectangle of fixed area ($h = \frac{S}{b}$).

If x doubles, y halves. If x is multiplied by 3, y is divided by 3. The product stays put.

The graph — a hyperbola

Take $k = 6$ and build a table:

x	1	2	3	6	−1	−2	−3	−6
y	6	3	2	1	−6	−3	−2	−1



The graph of y = 6/x. Two separate branches — never one connected curve.

The curve is called a **hyperbola**, and each half is a **branch**.

The properties

- **Domain:** every x except 0. The value $x = 0$ is not merely awkward — there is no point of the graph above or below the origin at all.
- **Range:** every y except 0. The curve never touches either axis.
- **Quadrants:** if $k > 0$ the branches lie in quadrants **I and III**; if $k < 0$, in **II and IV**.
- **Behaviour:** for $k > 0$ the function is decreasing on each branch separately — but *not* on the whole domain, because the two branches are not joined.
- **Asymptotes:** the curve comes arbitrarily close to both axes and never meets them.
- **Symmetry:** the graph is symmetric about the origin, and about the line $y = x$ (or $y = -x$ when $k < 0$).

SAY THIS OUT LOUD IN CLASS

"Decreasing on each branch" is not the same as "decreasing". Take $y = \frac{6}{x}$: at $x = -1$, $y = -6$; at $x = 1$, $y = 6$. The value went *up* as x went up — because we crossed the gap at $x = 0$.

Finding k

One point is enough. If the graph passes through $(3, 4)$ then $k = 3 \cdot 4 = 12$, so $y = \frac{12}{x}$.

And the test for whether a point lies on the graph is just as quick: multiply its coordinates and compare with k.

02 Worked examples

EXAMPLE 1 ***The graph of y = k/x passes through (−2, 5). Find k and describe the graph.***

1. $k = xy = (-2)(5) = -10$

   The product of the coordinates is k.

2. $y = -\dfrac{10}{x}$

   The rule.

3. $k < 0$

   so the branches lie in quadrants II and IV.

4. Domain $x \neq 0$, range $y \neq 0$.

   The axes are asymptotes.

ANSWER

$k = -10$; a hyperbola in quadrants II and IV.

EXAMPLE 2 ***Find where $y = 8/x$ meets $y = 2x$***

1. $$\frac{8}{x} = 2x$$

   Set the two rules equal.

2. $8 = 2x^2$

   Multiply both sides by x — legal, since $x \neq 0$.

3. $x^2 = 4$, so $x = 2$ or $x = -2$

   Both roots are permissible.

4. $x = 2 \Longrightarrow y = 4;\ x = -2 \Longrightarrow y = -4$

   Substitute back.

ANSWER

$(2, 4)$ and $(-2, -4)$

EXAMPLE 3 **A rectangle has area 36 cm². Express its height in terms of its base and name the graph.**

1. $b \cdot h = 36$

   Area of a rectangle.

2. $h = \dfrac{36}{b}$

   Make h the subject.

3. $k = 36 > 0$, and lengths are positive.

   So only the branch in quadrant I has meaning here.

4. A hyperbola — but only the part with $b > 0$ is drawn.

   Context can cut a graph down.

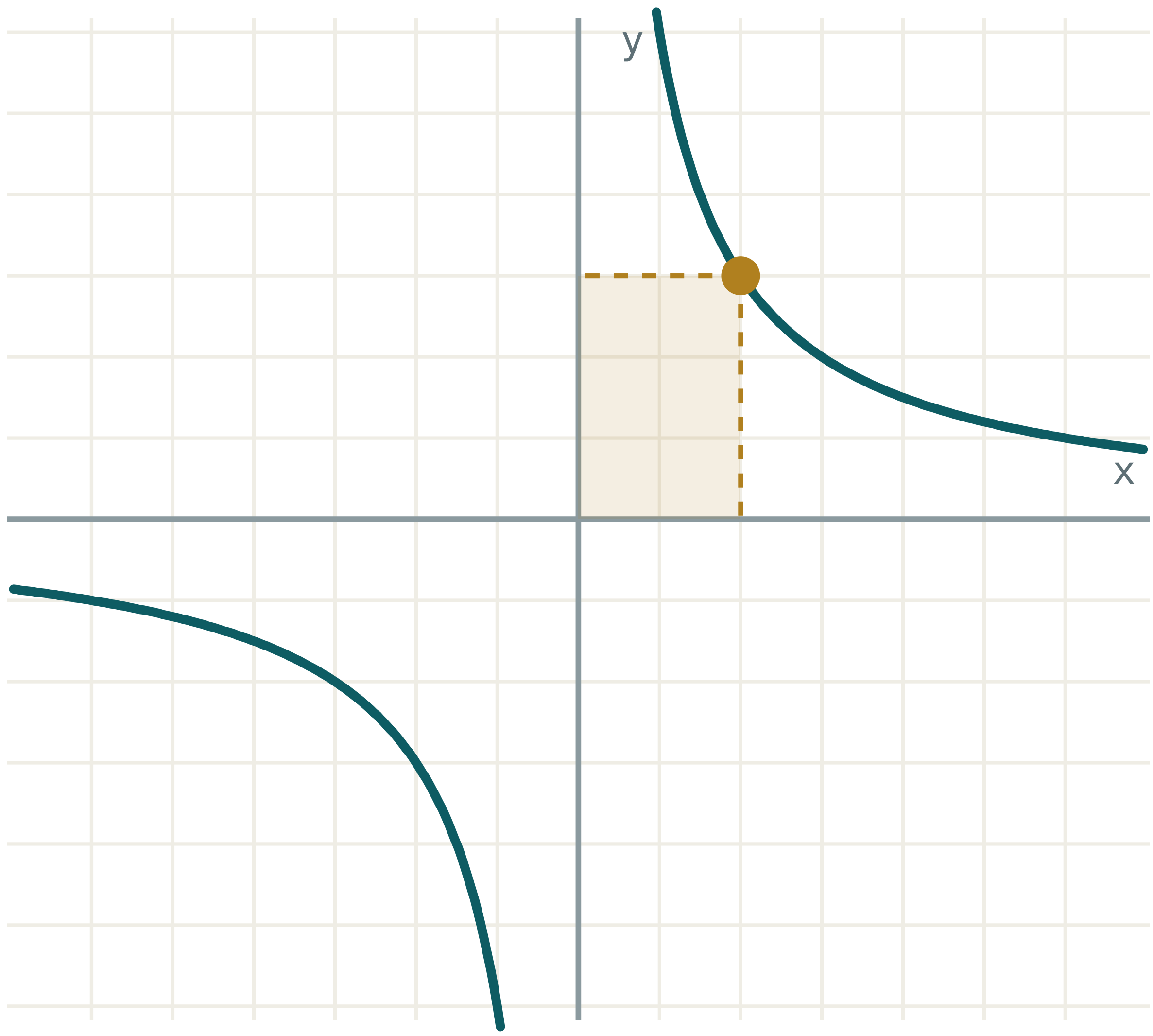
ANSWER

$h = \dfrac{36}{b}$; one branch of a hyperbola, $b > 0$

03 Interactive model

Drag k through zero and ask the class to predict which quadrants the branches jump to.

The graph of y = k ⁄ x



k **6**	x **2**	y = k/x **3**	x · y **6**	quadrants **I and III**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Function	Funksiya	Функция
Inverse proportion	Teskari proporsionallik	Обратная пропорциональность
Coefficient k	k koeffitsiyenti	Коэффициент k
Graph	Grafik	График
Hyperbola	Giperbola	Гипербола
Branch	Tarmoq	Ветвь
Domain	Aniqlanish sohasi	Область определения
Asymptote	Asimptota	Асимптота
Quadrant	Koordinata choragi	Координатная четверть
Symmetry	Simmetriya	Симметрия

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The graph of $y = \frac{7}{x}$ lies in quadrants:

- I and II
- I and III
- II and IV
- all four

If $y = \frac{k}{x}$ passes through $(3, 4)$ then k is:

- 12
- $\frac{3}{4}$
- 7
- $\frac{4}{3}$

For $y = \frac{k}{x}$, if x is multiplied by 4 then y is:

- multiplied by 4
- divided by 4
- unchanged
- multiplied by 16

The graph of $y = \frac{k}{x}$ meets the x-axis:

- at one point
- at two points
- at the origin
- never

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $y = \dfrac{12}{x}$. *Find y when x = 4*

   ANSWER $y = 3$

2. $y = \dfrac{6}{x}$. *Find y when x = −2*

   ANSWER $y = -3$

3. *Does the point (2, 5) lie on* $y = \dfrac{10}{x}$?

   ANSWER Yes — $2 \cdot 5 = 10$.

4. *In which quadrants does* $y = \dfrac{7}{x}$ *lie?*

   ANSWER I and III

5. *In which quadrants does* $y = -\dfrac{5}{x}$ *lie?*

   ANSWER II and IV

6. $y = \dfrac{8}{x}$. *Find x when y = 2*

   ANSWER $x = 4$

7. *Is* $y = \dfrac{3}{x}$ *defined at x = 0?*

   ANSWER No — $x = 0$ is not in the domain.

Medium · the standard question 7 PROBLEMS

1. *(3, 4) lies on $y = k/x$. Find k.*

   ANSWER $k = 12$

2. *(−2, 5) lies on $y = k/x$. Find k.*

   ANSWER $k = -10$

3. *For $y = k/x$, what happens to y when x doubles?*

   ANSWER y halves — the product xy is constant.

4. *Find k so that $y = k/x$ passes through (0.5, 8)*

   ANSWER $k = 4$

5. *Complete the table for $y = \dfrac{12}{x}$ at $x = 1, 2, 3, 4, 6, 12$*

   ANSWER $12, 6, 4, 3, 2, 1$

6. *(2, 9) and (a, 6) both lie on $y = k/x$. Find a.*

   ANSWER $k = 18$, so $a = 3$

7. *For $y = \dfrac{24}{x}$, list the positive integers x giving an integer y*

   ANSWER $1, 2, 3, 4, 6, 8, 12, 24$

Hard · stretch and reason

7 PROBLEMS

1. *Show that for $y = k/x$ the product xy is the same at every point*

   ANSWER $xy = x \cdot \frac{k}{x} = k$ for every permissible x.

2. *$y = k/x$ passes through $(-3, -4)$. Find k and the quadrants.*

   ANSWER $k = 12$; quadrants I and III.

3. *A rectangle has area 36. Write h in terms of b and name the graph.*

   ANSWER $h = \frac{36}{b}$; a hyperbola, $b > 0$ only.

4. *(m, n) lies on $y = k/x$. Show that (n, m) does too.*

   ANSWER $nm = mn = k$, so the second point satisfies the same rule.

5. *Solve $\frac{6}{x} = x$*

   ANSWER $x^2 = 6$, so $x = \pm\sqrt{6}$

6. *Find where $y = \frac{8}{x}$ meets $y = 2x$*

   ANSWER $(2, 4)$ and $(-2, -4)$

7. *Sketch $y = \frac{4}{x}$ and $y = -\frac{4}{x}$ together and describe the symmetry*

   ANSWER Each is the reflection of the other in either coordinate axis.

07 Homework

Homework — 6 problems

Algebra 8, §7, pp. 34–38. Draw the graph of question 4 on squared paper.

1. $y = \frac{15}{x}$. *Find y when $x = 3$, $x = 5$ and $x = -15$*
2. *$(4, 6)$ lies on $y = k/x$. Find k and write the rule.*

3. *In which quadrants does* $y = -\dfrac{9}{x}$ *lie? Explain in one sentence.*

4. *Draw the graph of* $y = \dfrac{4}{x}$ *for* x *from* -4 *to* 4*, using a table of values.*

5. A journey of 120 km takes t hours at speed v km/h. Write t in terms of v and find t when $v = 60$.

6. *Solve* $\dfrac{10}{x} = \dfrac{x}{10}$